Tridib Chakraborty

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Summary

I am a CSE graduate from East West University, majoring in Data Science with hands-on experience in AI/ML, LLMs, NLP, Computer Vision, Data Preprocessing, N8N AI- Automation. Skilled in custom dataset making, preprocessing, AI- Automation projects and end-to-end production grade AI powered project execution. I also attached the leading projects, including my capstone project & two research papers among the best.

Education

B.Sc in Computer Science and Engineering 2020-2025
East West University, Dhaka, Major: Data Science CGPA: 2.83/4.00

Projects

1. Capstone Project : A Multilingual Visual Question Answering App for cultural Heritage sites of Bangladesh GitHub
Technologies: Python, PyTorch, Hugging Face Transformers, BLIP, Streamlit

- Curated and cleaned a multilingual VQA dataset for Bangladeshi cultural heritage sites by aligning heritage images with corresponding textual question-answer annotations.
- Applied text augmentation techniques, including word swapping and sentence swapping, to increase question diversity and improve dataset variability.
- Processed image-question pairs into model-ready tensors for vision-language model training and inference.
- Fine-tuned the BLIP (Bootstrapped Language-Image Pre-training) model on the curated VQA dataset to generate answers from heritage images and text-based questions.
- Developed a Streamlit-based VQA application for real-time image-question input and AI-generated responses.

2. Production-Grade Multimodal VQA System (BLIP + LLaMA 3.3 70B) for Understanding household things and products in shops GitHub
Technologies: PyTorch, Hugging Face Transformers, BLIP, FastAPI, React.js, Groq, LLaMA 3.3 70B

- Fine-tuned the BLIP vision-language model on 1,400+ image–question–answer pairs for domain-specific Visual Question Answering (VQA).
- Built a multimodal inference pipeline combining image embeddings and text queries to generate contextual answers from visual inputs.
- Developed FastAPI REST APIs for model inference and integrated the fine-tuned BLIP model with a React.js frontend for real-time interaction.
- Integrated LLaMA 3.3 70B through Groq API to refine raw BLIP outputs into more natural and contextually relevant responses.
- Combined the fine-tuned BLIP + LLaMA 3.3 70B (VLM + LLM) models into a two-stage pipeline for practical vision-language application deployment.

3. AI-Powered Customer Review Email Automation
Technologies: n8n, Gmail API, GPT API, Vector Database

- Integrated Gmail API with n8n to fetch and analyze recent customer review emails.
- Built an LLM-powered workflow using GPT API and prompt engineering for context-aware response generation.
- Implemented vector-based context retrieval to improve the relevance and consistency of generated email responses.

4. Waste Detection and Classification using YOLOv8 Kaggle
Technologies: Python, PyTorch, YOLOv8, Roboflow

- Preprocessed 4,959 annotated waste images across 5 classes using image filtering and augmentation techniques.
- Trained and evaluated YOLOv8n, YOLOv8s, and YOLOv8m models for waste detection and classification.
- Achieved 89.6% precision and 81.33% accuracy with YOLOv8n it also maintaining the fastest inference performance.
- Identify the plastic bottle and other wastes in real time using camera images.

5. Movie Recommendation System GitHub
Technologies: Python, TMDB API, Streamlit, TF-IDF, Cosine Similarity

- First of all Cleaned and processed 5,000+ TMDB movie records, combining relevant content features for recommendation.
- I Used TF-IDF vectorization and cosine similarity to calculate movie-to-movie similarity based on content features.
- Built a Streamlit web application with TMDB API integration to provide interactive movie recommendations.

PAPER AND PUBLICATIONS

1. Multiple Pox and Skin Diseases Classification using Transfer Learning Approach
Technologies: Python, PyTorch, DenseNet201, MobileNet, Image Augmentation
DOI: 10.1007/978-3-032-25949-3_16

- Curated and preprocessed a dataset of 6,756 multi-class skin disease images, applying image filtering and augmentation techniques..
- Fine-tuned DenseNet201 , MobileNet and others 3 CNN models using transfer learning and compare for best performance for automated multi-class skin disease classification.
- DenseNet201 achieved 98.39% accuracy and the study was published at ICDMIS 2025..

2. Colorectal Cancer Classification Using Transfer Learning
Technologies: Python, PyTorch, DenseNet201, SENet, InceptionResNetV2, NASNet
DOI: 10.1007/978-3-032-25949-3_30

- Prepared and standardized a dataset of 8,750 colorectal cancer histopathology images across 11 classes using preprocessing and image augmentation techniques.
- Evaluated multiple transfer learning architectures, including DenseNet201, SENet, InceptionResNetV2, NASNetMobile, and NASNetLarge.
- DenseNet201 achieved the best performance with 94.46% accuracy, demonstrating effective feature extraction for multi-class colorectal cancer classification.

Technical Skills

1. **Programming:**
   Python, SQL, JavaScript, C, Java
2. **Machine Learning & Deep Learning:**
   PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, **Transfer Learning**, **Model Fine-Tuning**, Model Evaluation
3. **Generative AI & LLM:**
   LLMs, **Generative AI**, NLP, **Transformer Models**, **Hugging Face Transformers**, Prompt Engineering, **RAG**, Embeddings, Vector Search, LangChain, LLM API Integration
4. **Computer Vision:**
   YOLOv8, **BLIP**, Image Classification, Object Detection, **Visual Question Answering (VQA)**, Image Augmentation, OpenCV
5. **AI Engineering & APIs:**
   FastAPI, **RESTful APIs**, Streamlit, React.js, n8n, OpenAI/GPT API, Groq API, Real-Time Inference
6. **Data & Vector Databases:**
   Pandas, NumPy, PostgreSQL, MySQL, **Pinecone**, ChromaDB, Vector Databases
7. **MLOps & Deployment:**
   Docker, ML Pipelines, **Model Deployment**, AWS, Azure, GCP
8. **Developer & Research Tools:**
   Git, GitHub, VS Code, PyCharm, Google Colab, Kaggle, Roboflow, Overleaf

Certifications & Achievements

DataCamp — Data Science & Machine Learning Courses
Completed courses with certificates:

- **Intermediate Regression with statsmodels in Python**
- **Introduction to Regression with statsmodels in Python**
- **Intermediate Data Visualization with Seaborn**
- **Time Series Analysis in Python**
- **Unsupervised Learning in Python**
- **Machine Learning with Tree-Based Models in Python**
- **Feature Engineering for Machine Learning in Python**

References

Dr. Mohammad Rifat Ahmad Rashid
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